School of Computer & Systems Sciences

CS – 408 Computer Networks

MCA

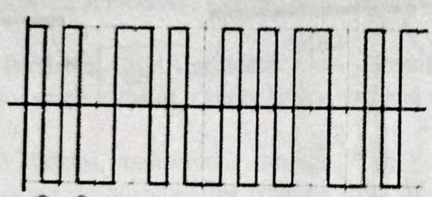
Mid Semester Examination - 2025

**Time: 120 Minutes** **Max. Marks: 30**

***Note: Attempts all questions. Q1 to Q5 carry 2 marks each and Q6 to Q9 carry 5 marks each.***

**Q1.** Which of the OSI layers handles each of the following:
(a) Determining which route through the subnet to use.
(b) Dividing the transmitted bit stream into frames.
(c) Concern with the syntax and semantics of the information.
(d) Check-pointing long transmission to allow them to continue from where they ware after a crash.

**Q2.** In the below figure 1, the Differential Manchester encoding of data stream is given. What is the data stream?



0 0 1 0 0 1 1 1 1 0 1 1

Fig. 1

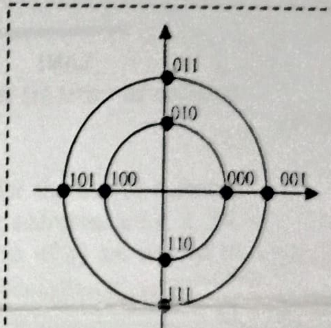


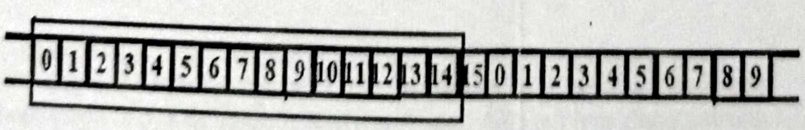

Fig. 2

**Q3.** Draw the time domain with the given constellation in fig 2 for a data stream 101000010111001110011. Assume that baud interval is 2 cycle's time of the wave.

**Q4.** The wireless LANs that we studied used protocols such as MACA instead of using CSMA/CD. Under what conditions, if any, would it be possible to use CSMA/CD instead?

**Q5.** Calculate the channel efficiency of Binary Countdown protocol if we åssume that there are 800 stations contending for the same channel and data frame consisting of 512 bits.

**Q6.** The sender windows for system uses Go-Back-n ARO are given as:




(i) Show the window after the sender has sent packets 0 to 12 and has received ACK7.
(ii) Show the window after the sender has sent packets 0 to 14 hand has received NAK6.
(iii) Show the window after the sender has sent packets 0 to 14, no acknowledgement has received, and timeout has expired.
(iv) The receiver has sent ACK6 and ACK9 but ACK6 is lost and no timeout up to ACK9 at sender. Show the sender window.

P.T.O.

**Q7.** Sixteen-bit message are transmitted using a Hamming code. How many check bits are needed to ensure that the receiver can detect and correct single bit errors? Show the bit pattern transmitted for the message 1101001100110101. Assume that even parity is used in the Hamming code.

**Q8.** A modified NRZ code known as enhanced-NRZ (E-NRZ). E-NRZ encoding entails separating the NRZ-L data stream into 7-bit words; inverting bits 2, 3, 6 and 7; and adding one parity bit to each word. The parity bit is chosen to make the total number of 1s in the 8-bit word an odd count. What are the advantages of E-NRZ over NRZ-L? Any disadvantages?

**Q9.** Six stations (S1 to S6) are connected to an extended LAN through transparent bridges (B1 and B2) as show in figure in 3 below. Initially, the forwarding (bridge) tables are empty. Suppose the following stations transmit frames: S2 transmits to S1, S5 transmits to S4, S3 transmits to S5, S1 transmits to S2 and S6 transmits to S5. Fill in the forwarding tables with appropriate entries after the frame have been completely transmitted.

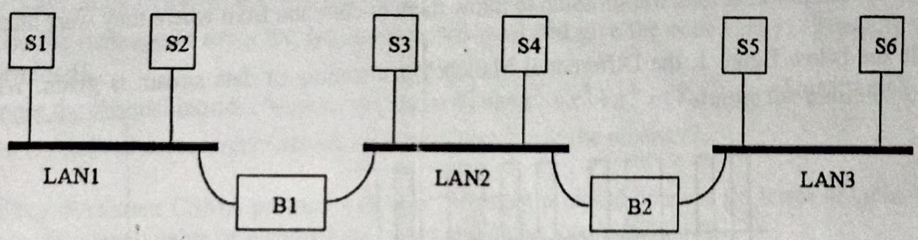



Fig. 3

P.T.O.

**School of Computer & Systems Sciences**
**Computer Networks**
**MCA II – Semester**
**End Semester Examination-2025**

**Time: 3 Hours** **Max. Marks: 50**

***Note*: *Attempt all questions***

**Q1.** The following encoding methods for framing are used in a data link protocol: *A*: 010100111; *B*: 11100011; *C*: *11001111*, *FLAG*: 01111110; *ESC*: 11100000. Show the bit sequence transmitted (in binary) for the five-character frame: *A C B ESC FLAG* when the following methods are used: ***(5)***
(i) Flag byte with byte stuffing
(ii) Starting and ending flag bytes, with bit stuffing.

**Q2.** Suppose we want to transmit the 8-bit message $M$ = *11101011* and protect it from errors using the generator polynomial $P(x)=x^5+x^4+x+1$. ***(5+5)***
(i) Encode the message $M$ using the generator polynomial and give the codeword (*T*: Frame to be transmitted).
(ii) Suppose the channel introduces an error pattern $E(x)=x^9+x^6+x^3$ in *T* during the transmission. What is received at receiver? Can the error be detectable at the receiver?

**Q3.** **(a)** What is *p*-Persistent CSMA protocol? How *p*-Persistent protocol behaves (in terms of collision and delay) for small value of *p* under heavy load and under low load? ***(5+5)***

(b) Consider all stations, numbered *1* through $2^n$ ($n \geq 2$), are contending for the use of a shared channel by using the adaptive tree walk protocol. If all the station whose addresses are *4, 8, 16, 32, 64....* suddenly becomes ready at once, how many bit slots (in terms of *n*) are needed to resolve the contention?

**Q4. (a)** Consider the subnet of Fig 1. Distance vector routing is used, and the routing tables are established for routers *A, B, C, D, E*, and *F* (as given in Table 1): and then the *C-E* link fails. Give ***(5+5)***
(i) The routing tables of *A, B, D*, and *F* after *C* and *E* have reported the news to their neighbors.
(ii) The routing tables of *A* and *D* after their next mutual exchange.
(iii) The routing table of *C* after *A* exchange with it.

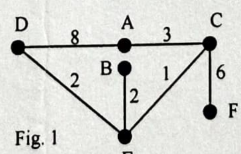
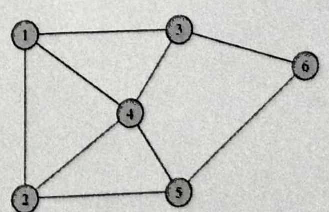


Fig. 1

**Table-1**

| Routing table at router | Cost to reach router A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| A | 0 | 6 | 3 | 6 | 4 | 9 |
| B | 6 | 0 | 3 | 4 | 2 | 9 |
| C | 3 | 3 | 0 | 3 | 1 | 6 |
| D | 6 | 4 | 3 | 0 | 2 | 9 |
| E | 4 | 2 | 1 | 2 | 0 | 7 |
| F | 9 | 9 | 6 | 9 | 7 | 0 |

(b) Explain the flooding. Consider the network as show in the figure, using flooding as the routing algorithm. If a packet sent by *1* to 6 has a maximum hope count of *3*. List all the routes it will take. Also tell how many packets generated? Assume, no duplicate is discarded.




**Q5. (a)** Identify the address class of the following IP addresses: 200.58.20.165, 128.167.23.20, 16.196.128.50, 150.156.10.10 and 250.10.24.96 ***(5+ 5)***
**(b)** A small organization has a class C address for 7 networks each with 24 hosts. What is an appropriate subnet mask?

**Q6.** Suppose a router receives an IP packet containing 1300 data bytes and has to forward the packet to a network with maximum transmission unit of 300 bytes. Assume that the IP header is 20 bytes long. Show the fragments that the router creates and specify the relevant values in each fragment header (i.e. total length, fragment offset, and more bit). ***(5)***